Continuous Random Variables

6.1 Probability density functions

Definition of Probability Density Function (PDF)

A.K.A density function, a conutinuous version of the PMF, and this function should always output a non-negative real value

The probaility is $P(X \in A) = \int_A f(x)dx$

(a) $F(t) = \int_{-\infty}^{t} f(x)dx$

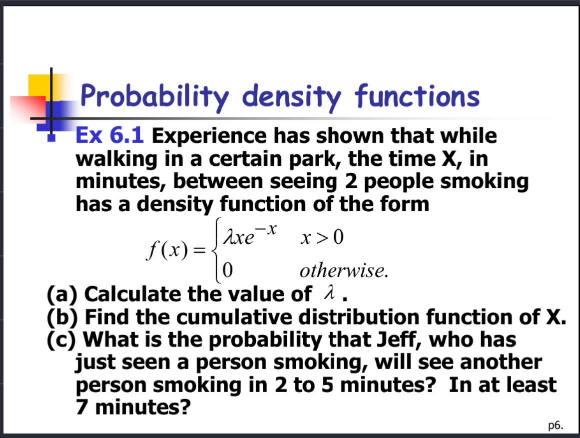
(b) $\int_{-\infty}^{\infty} f(x)dx = 1$

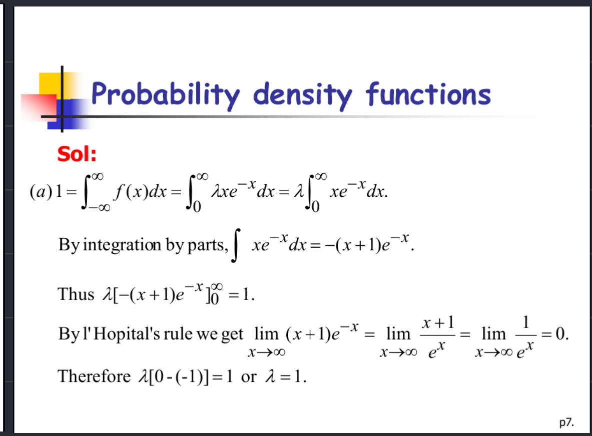
(c) If f is continuous, then F'(x) = f(x), 即使f不連續，在f連續的部分 F'(x) = f(x)

(d) For real number $a \leq b$, $P(a \leq X \leq b) = \int_a^b f(t)dt$

(e) $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = \int_a^b f(t)dt$

表示在連續分布中 P(x) = 0





6.2 Density function of a function of a random variable

How to ...

Let $Y = h(x) = t^2$, then $P(x < Y) = P(x < t^2) = P(-\sqrt{x} \leq t \leq \sqrt{x})$

6.3 Expectations and variances

Excepted value of Continuous R. V. Definition

A continuous R.V. with P.D.F f(x), has the mean/excepted value E(X):

$$E(X) = \int_{-\infty}^{\infty} x \times f(x)dx$$

TODO: p.28

Cauchy random variable

這邊只是證明積分後機率加總為 1，可以不用被證明過程

Theorem 6.2 For any continuous R. V. X with distribution function F and density function f,

$$E(X) = \int_0^\infty [1 - F(t)]dt - \int_0^\infty F(-t)dt.$$

$$E(X) = \int_0^\infty P(X > t)dt - \int_0^\infty P(X \le -t)dt.$$

如果對此不感興趣，可以直接將 F(X) 微分得到 f(x) 後，直接對其作積分 (Def 1)

Theorem 6.3

Let X be a continuous R.V. with density function f(x);

then for any function h(x): R → R

$$E[h(x)] = \int_{-\infty}^{\infty} h(x) \cdot f(x)dx$$

且這邊的期望值也遵守線性三條

Definition of Varience

$$Var(x) = E(X^2) - (E(X))^2$$